



Sharing Research Software

Reproducible, Citable, and Discoverable

Tim Dennis

Data Science Center, UCLA Library

2026-09-23

Welcome!

UC Carpentries Workshop Series · Fall 2026

We'll begin shortly. Grab the collaborative notes and settle in.

pad.carpentries.org/2026-fall-uc-sharing-research-software · ucospo.net/research-software-citable-discoverable

Code of Conduct

We agree to

- Use **welcoming and inclusive** language
- Be **respectful** of different viewpoints and experiences
- Accept **constructive criticism** gracefully
- Show **courtesy and respect** toward everyone here

If something's wrong

- Tell an **instructor** or the **host** — chat DM or email, privately
- Or report to the **CoC Committee**:
coc@carpentries.org
- All reports are **kept confidential** and taken seriously

Full text & report form: docs.carpentries.org/policies/coc

How we learn together

Our pedagogy

- **Live coding:** we learn by doing together, and mistakes are teaching moments, not detours
- **Mental models:** new skills land when we connect them to work you already do
- **No overload:** a few key concepts done well beats racing through everything

Managing expectations

- **Practical competence:** skills you can use in your role right away, not exhaustive coverage
- **Collaborative pace:** we move as a group, and your questions help set the speed
- **Mistakes are expected:** coding is troubleshooting and persistence, for all of us

Software Is the Missing Scholarly Output

The question that starts this lesson

“The journal accepted my paper, but they want a citable version of the code, with a DOI. How do I do that?”

Ten years ago, the same question arrived about **datasets** — and the library built the answer: a persistent identifier, a catalog record, a citation format, a preservation copy.

Software is the third thing modern research runs on. Mostly, it hasn't gotten that treatment yet.

Who this lesson is for

You write research software

...and want to make your own code citable, reusable, and discoverable.

You support people who do

...as a librarian, OSPO member, or research-support specialist advising others.

The hands-on steps are the same either way. Watch for **Supporting others** call-outs — they translate each technical step into advising and consultation work.

How common is the software citation gap?

Research software is fundamental to modern scholarship, but challenges remain around its discoverability, reproducibility, and sustainability.

59.5%

of 3.2 million PubMed Central publications mentioned at least one software item

38%

of 2020 publications gave enough detail to **uniquely identify** the software used

Barker et al. (2022) doi:10.1038/s41597-022-01710-x · Schindler et al. (2022) doi:10.7717/peerj-cs.835

A GitHub URL isn't a citation

The wrong way

“We used the analysis script from `https://github.com/UC-OSP0-Network/software-demo` ”

- URLs break (link rot)
- No version specified — which run produced which result?
- No formal credit; the author's name isn't even visible
- Cannot appear in citation metrics

The right way

Phan, L., Padilla, K., Otsuji, R., & Dennis, T. (2026). *Biodiversity Analysis Toolkit* (v0.1.0). Zenodo. <https://doi.org/10.5281/zenodo.123456>

- DOI is permanent — survives username changes, repo moves, even GitHub disappearing
- Version is explicit, so someone can reproduce the exact run
- Author gets formal credit and citation metrics

The fix is a `CITATION.cff` file and a DOI — both of which you'll create in this workshop.

Software has a durability problem

The Bus Factor

The number of people who need to be “hit by a bus” before a project becomes unmaintainable.
For most research code, the answer is one.

How many people would have to leave before this software becomes unrecoverable? A library would never accept a collection with a bus factor of one — that’s exactly what an un-deposited, un-documented repository is.

Can anyone actually run it?

There's a third problem — the one researchers feel first: software that only runs on its author's machine.

```
1 $ python src/analysis.py
2
3 ModuleNotFoundError: No module named 'numpy'
```

The core lesson makes software **citable and discoverable**. The **optional** pixi episode at the end makes it **runnable**, too.

The fix: FAIR4RS principles

F — Findable DOI, CITATION.cff, metadata

A — Accessible Public GitHub, archived on Zenodo

I — Interoperable Standard formats, documented dependencies

R — Reusable LICENSE, README, environment files

What you're doing today

-  **License** — so others can legally reuse your work
-  **Citation** (CITATION.cff + DOI) — so you get academic credit
-  **README & metadata** — so people can find and understand it
-  **Environment** (*optional*) — so it also runs on any machine

Before → after

Before (main)

- ❌ No LICENSE
- ❌ No citation information
- ❌ No DOI
- ❌ Minimal README
- ❌ No community or metadata files

After (05-add-metadata)

- ✅ LICENSE (BSD-3-Clause)
- ✅ CITATION.cff
- ✅ Complete README
- ✅ CONTRIBUTING.md,
CODE_OF_CONDUCT.md
- ✅ .zenodo.json, DOI badge

This is what makes software Findable, Accessible, Interoperable, and Reusable.

Real-world example: Spack

Used by national labs, universities, and supercomputing centers: github.com/spack/spack

-  README with clear description and getting-started guide
-  LICENSE (MIT and Apache-2.0)
-  CITATION.cff file
-  Documentation
-  Code of conduct and contributing guide

What we'll build together

Starting from the bare `main` state in your fork, progressively:

1. Sharing research software (*exploring the starting repository*)
2. Adding an open-source license
3. Creating a CITATION.cff file
4. Minting a DOI and creating a release
5. Improving metadata and discoverability
6. Managing environments with pixi (*optional*)
7. Wrap-up and reflection

View-only reference branches (`02-add-license` , `03-add-citation` , `04-create-release` , `05-add-metadata`) let you check your work at any point.

Sharing Research Software Effectively

Present, but not yet usable

Every archivist knows the unprocessed accession: boxes that made it into the building but not into the collection — technically held, practically invisible.

Most research software on GitHub is in exactly that state. **Public is not the same as shared.**

An open repository with no license, no citation, no usable description is like leaving the reading room unlocked and calling it access: people can technically get in, and nothing helps them once they do.

Fork the demo repository

Every hands-on step happens in **your own fork**, not the shared demo repo.

1. Open `github.com/UC-OSP0-Network/software-demo`
2. Click **Fork** → **Create fork** (defaults are fine — `main` branch only)
3. Confirm the top-left reads `YOUR-USERNAME/software-demo` , not `UC-OSP0-Network/software-demo`

Reference branches are there to check your work, never to build on. Full setup details: ucospo.net/research-software-citable-discoverable/learners/setup

The starting point

```
1 software-demo/  
2 |— README.md  
3 |— requirements.txt  
4 |— src/  
5   |— analysis.py
```

This repository intentionally *lacks* many elements of good research software practice. By the end of the lesson, it will include licensing, citation metadata, improved discoverability, and versioning.

Supporting others: a four-question triage

When a researcher brings you their repo, you're doing a fast triage, not a full code review — a reference interview, not a dissertation defense.

1. **Rights** — is there a license? Can this legally be reused at all?
2. **Credit** — can a stranger tell who made it and how to cite it?
3. **Runnable** — is there any record of what it takes to run?
4. **Findable** — would anyone outside the lab ever discover it?

The goal is the **single highest-value next step**, not fixing everything.

Try it: inventory the demo repository

Open your fork of **software-demo**. Spend 2 minutes, then answer:

- Could you run this code today? What's missing?
- Would you know who created it, or how to credit them?
- Could you legally reuse or modify it?

We'll address every one of these gaps together over the next episodes.

Key points

- Software is a research output the scholarly system has mostly failed to catch
- A GitHub URL is not a citation — DOIs are permanent and version-specific, URLs rot
- Undeposited software has a bus-factor problem; forges do shut down
- FAIR4RS (Findable, Accessible, Interoperable, Reusable) is the framework
- Public repositories increase findability, reuse potential, and citation credit
- You'll build up your own fork step by step, episode by episode